Learning Energy-Efficient Uplink Access with Asynchronous Packet Deadlines

Author information to be supplied

***Abstract*—Periodic devices can save transmit energy by spreading service over several slots, but overlapping transmissions can erase this benefit in shared uplinks. We study distributed subchannel, power, and silence decisions for asynchronous periodic packets under per-device deadline requirements. Devices use local packet state, delayed public received-power measurements, and private service feedback. A shared recurrent policy is trained using multi-agent proximal policy optimization and compared with calibrated local rules and centralized zero-loss references. Across 200 held-out instances per network size, three training seeds, and two executions per instance, the fixed-penalty policy reduces mean normalized transmit energy by 36.6%, 56.7%, and 70.5% relative to the strongest evaluated feasible local rule for 6, 8, and 12 devices sharing four subchannels. Its feasible-episode fraction is at least 0.984. With four devices, energy is about 0.7% above the exact zero-loss reference. On a separate 200-instance confirmation set with eight devices, energy is 59.8% lower than an XR-SBPC-style discrete soft actor–critic adaptation using the same energy reward and execution information. The learned policy combines low-power transmission with selective silence and exhibits lower co-channel overlap and interference loss. These results demonstrate the energy benefits of learned distributed access for asynchronous periodic traffic.**

***Index Terms*—distributed access, periodic traffic, deadline constraints, power control, multi-agent reinforcement learning**

I. Introduction

Periodic sensing and control traffic poses an energy–reliability tradeoff in shared uplinks. A device can lower its power and accumulate service across several slots before a deadline, as motivated by classical energy-efficient packet scheduling on single and multiple-access links [1], [2]. However, extending transmissions increases the opportunities for interference. With heterogeneous periods and unaligned arrival phases, distributed devices must decide which subchannel and power to use, and when to remain silent despite a pending packet. The resulting energy–reliability tradeoff calls for per-device delivery constraints and energy references that account for interference.

Deadline-constrained scheduling characterizes per-client delivery requirements [3] and periodic traffic with general offsets [4], but relies on a central scheduler; delay-constrained grant-free access has been treated with distributed pilot and power selection [5]. Recurrent policies with local acknowledgments have been used for distributed spectrum access [6], and vehicular MARL jointly selects subbands and powers under payload deadlines [7]. Reliability-constrained power control for periodic mission-critical transmissions [8] and recurrent spectrum/power allocation in multicell networks [9] further motivate this setting. The close XR-SBPC study [10] combines periodic traffic with offsets, accumulated Shannon service, and subband–power–silence actions. It optimizes reliability in downlink subnetworks, using neighboring actions and current link gains. We study transmit-energy minimization with per-device delivery requirements in a common-receiver uplink, using only local packet history and delayed AP feedback.

We develop a distributed controller with a shared, subchannel-equivariant recurrent actor trained by multi-agent proximal policy optimization (MAPPO) [11]. A fixed violation penalty is compared with local dual adaptation inspired by state-augmented constrained reinforcement learning [12]. Our contributions are:

1) A causal distributed access formulation for asynchronous periodic packets, with accumulated service and joint channel, power, and silence decisions.
2) Centralized zero-loss energy references for the same discrete-power model: an exact solution when $N \le K$, and lower and feasible upper references when devices outnumber subchannels.
3) Held-out evaluation against calibrated local rules and an XR-SBPC-style SAC adaptation. Multiple seeds, paired intervals, and a fresh confirmation set quantify reward effects within SAC and the energy advantage of the MAPPO scheme. Transmission statistics reveal lower power and reduced harmful overlap in the learned access patterns.

II. System Model and Control Objective

A. Periodic uplink and accumulated service

Figure 1 illustrates an access point (AP) receiving uplink packets from N devices over K orthogonal subchannels. Slots have duration τ, and each subchannel has bandwidth B. Device i generates a packet of L_i bits at $a_{i,m} = \varphi_i + mP_i$, with deadline $d_{i,m} = a_{i,m} + P_i$. The phase φ_i is drawn independently and uniformly from $\{0, \dots, P_i - 1\}$. Physical slots are synchronized; asynchronous traffic refers to the packet-arrival phases. At most one packet is pending per device.

In slot t, device i selects $a_i(t) \in \{\mathsf{OFF}\} \cup (\{1, \dots, K\} \times \mathcal{P})$, where $\mathcal{P}$ is the positive discrete-power menu. Let $k_i(t)$ and $p_i(t)$ denote its chosen subchannel and power, with $p_i(t) = 0$ for OFF. The noise power per subchannel is σ^2. The AP decodes each device's signal by treating interference as noise. For an active device,

$$\gamma_i(t) = \frac{p_i(t) g_i}{\sigma^2 + \sum_{j \neq i : k_j(t) = k_i(t)} p_j(t) g_j}, \tag{1}$$

$$b_i(t) = \tau B \log_2(1 + \gamma_i(t)), \qquad e_i(t) = \tau p_i(t). \tag{2}$$

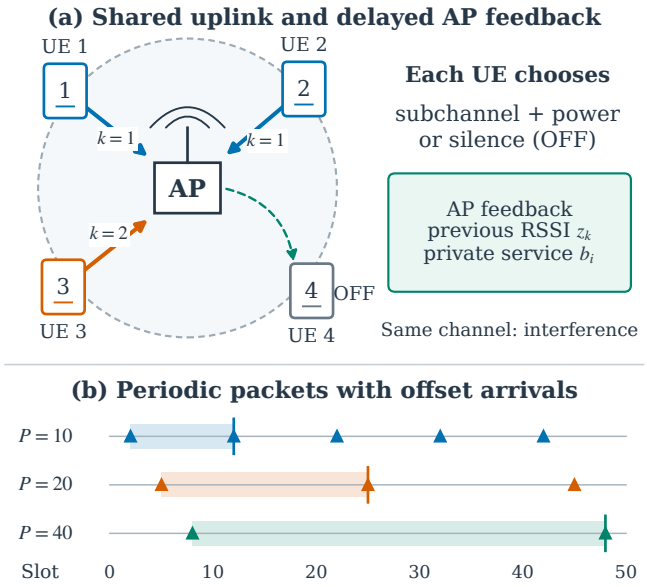


Fig. 1. Periodic uplink access. Solid arrows illustrate transmissions; UEs 1 and 2 reuse a subchannel and interfere at the AP. Dashed feedback is received even during silence. Shading marks the first packet window in each illustrative traffic class; triangles denote arrivals and vertical bars its deadline. Physical slots are synchronized, while arrival phases differ.

The gain g_i is static and identical across subchannels within an instance; subchannel selection controls co-channel interference. For silence, $b_i(t) = e_i(t) = 0$.

We use an ideal accumulated-mutual-information model, motivated by incremental redundancy hybrid ARQ [13]. A packet completes once the sum of its slot services reaches L_i. Even if its final transmission provides excess service, a device occupies and pays for the whole slot. At each slot boundary, the environment first applies service, then settles expiring packets, and finally releases new packets. Unfinished bits are discarded at expiry. A device without a pending packet must select OFF but continues receiving observations. We measure radiated transmit energy using (2).

B. Causal execution information

At a slot's start, device i knows its remaining bits $s_i(t)$, remaining time $D_i(t)$, its packet size and period, and its preceding action and outcome. The AP supplies the previous slot's per-subchannel received power $z_k(t-1)$, including noise, and the device's private service feedback $b_i(t-1)$. It supplies no scheduling decision. Devices do not exchange messages, and the policy receives no device identifier, global slot index, topology, or other devices' queues, phases, gains, and actions.

These observations contain useful channel information. After transmitting at power p on subchannel k, a device can infer

$$\widehat{\gamma} = 2^{b/(\tau B)} - 1, \qquad \widehat{I} = \frac{z_k}{1+\widehat{\gamma}}, \qquad \widehat{g}_i = \frac{z_k - \widehat{I}}{p}. \tag{3}$$

The gain estimate is retained across packets. Exact feedback and a static channel make this estimate exact after a transmission in the ideal model; the remaining uncertainty includes other devices' future actions. All distributed baselines use the same permitted observation stream.

C. Energy and per-device reliability

Let $c_{i,m} = 1$ if packet m of device i expires unfinished. For a fixed instance, the policy objective is

$$\min_{\pi} \quad J(\pi) = \limsup_{T\to\infty} \frac{1}{NT}\mathbb{E}_\pi\left[\sum_{t=0}^{T-1}\sum_i e_i(t)\right], \tag{4}$$

$$\text{s.t.} \quad \limsup_{M\to\infty} \frac{1}{M}\mathbb{E}_\pi\left[\sum_{m=0}^{M-1} c_{i,m}\right] \le \epsilon, \quad \forall i.$$

The per-device constraints assign the same violation allowance to every device. We train the policy across random instances using a penalized reward and each device's local history. Section V defines the finite-episode reliability criterion used for evaluation.

III. Centralized Energy References

We compute reference energies with full instance information and the same power menu, slot duration, and service model. These references require *every packet* to meet its deadline. The subscript 0 below denotes this zero-loss requirement, distinct from the $\epsilon = 0.05$ allowance for learned policies.

A. Interference-free per-packet solution

For device i, define the isolated service and energy of level ℓ as $b_{i\ell} = \tau B \log_2(1 + p_\ell g_i/\sigma^2)$ and $e_\ell = \tau p_\ell$. Its minimum isolated packet energy is the small integer problem

$$q_i = \min_{\boldsymbol{n}\in\mathbb{Z}_+^{|\mathcal{P}|}} \left\{ \sum_\ell n_\ell e_\ell : \sum_\ell n_\ell b_{i\ell} \ge L_i, \quad \sum_\ell n_\ell \le P_i \right\}. \tag{5}$$

Static channels make transmission slots interchangeable for this isolated problem. We enumerate the counts of all but the lowest positive power and fill the lowest level minimally. With five levels and periods no larger than 40, this yields an exact discrete solution. For a fixed evaluated packet set with M_i packets from device i, let $L_0 = \sum_i M_i q_i$.

Proposition 1. *For the zero-loss centralized scheduling problem, L_0 is a lower bound on energy. If $N \le K$, assigning one subchannel to each device attains L_0.*

Proof. For fixed power, service in (1) cannot exceed isolated service. Every successfully delivered packet therefore costs at least q_i; summing over packets gives the lower bound. If $N \le K$, distinct subchannels eliminate interference. Repeating each device's solution of (5) within its nonoverlapping packet windows attains the sum, independently of arrival phases. □

B. Cyclic orthogonal witness at higher loads

Let $C = \mathrm{lcm}(P_1, \ldots, P_N)$, which is 40 slots in our setup. For every cyclic packet window w of device $i(w)$, choose integer power-level counts $n_{w\ell}$. Let $y_{it} \in [0,1]$ indicate slot

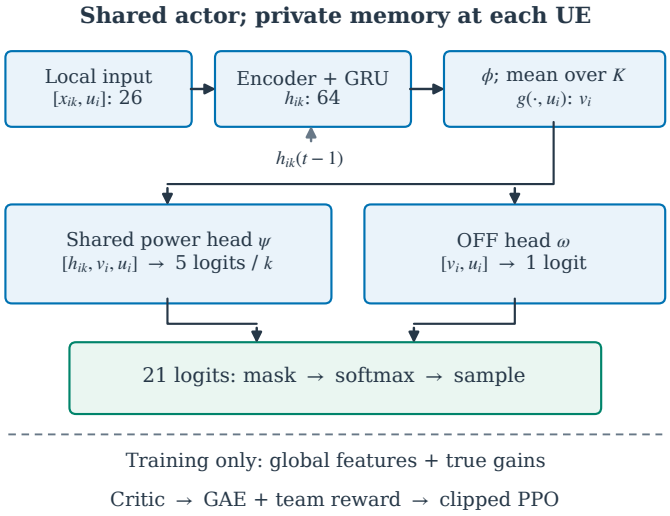


Fig. 2. Shared recurrent actor and centralized training. Each device keeps private per-subchannel hidden states. The encoder, GRU, and power head share weights over subchannels. The critic is used only during training.

assignment over one cycle. An orthogonal schedule is obtained from

$$\begin{aligned} \min_{n,y} \quad & \sum_{w,\ell} e_\ell n_{w\ell} \\ \text{s.t.} \quad & \sum_{t\in w} y_{i(w)t} = \sum_{\ell} n_{w\ell}, \quad \forall w, \\ & \sum_{i} y_{it} \le K, \quad \forall t, \\ & \sum_{\ell} b_{i(w)\ell} n_{w\ell} \ge L_{i(w)}, \quad \forall w, \\ & n_{w\ell} \in \mathbb{Z}_+, \quad 0 \le y_{it} \le 1. \end{aligned} \tag{6}$$

For fixed integer counts, the assignment constraints form a capacitated bipartite flow problem with integral capacities. An integral assignment can therefore be recovered by maximum flow. Each slot's active devices are assigned distinct subchannels, and the selected powers are placed within their packet windows. Repeating this schedule gives a feasible zero-loss witness, whose energy U_0 is measured by replay in the communication environment. We verify packet completion and energy accounting for every reference instance.

Consequently $L_0 \le E_0^\star \le U_0$ for the same zero-loss packet set, with equality when $N \le K$. We report E/L_0 and E/U_0 to quantify learned-policy energy relative to these centralized references.

IV. Learning Distributed Access

A. Shared recurrent actor

Figure 2 shows the actor and centralized training pathway. All devices share one actor but maintain private recurrent states. Its subchannel tokens contain delayed RSSI, inferred interference, indicators of the device's preceding transmission and power level, and estimated achievable service at each power divided by required service s_i/D_i. The context contains normalized remaining bits and time, the period, previous outcome and service, and the remembered gain with a validity indicator. Logarithmic scaling and clipping are applied to the rate ratios and physical quantities.

A shared encoder and gated recurrent unit (GRU) process each subchannel token $x_{ik}(t)$ together with device context $u_i(t)$:

$$h_{ik}(t) = \mathrm{GRU}_\theta(f_\theta(x_{ik}(t), u_i(t)), h_{ik}(t-1)), \tag{7}$$

$$v_i(t) = g_\theta\left(K^{-1}\sum_k \phi_\theta(h_{ik}(t)), u_i(t)\right). \tag{8}$$

A shared head maps (h_{ik}, v_i, u_i) to the power logits for subchannel k; a separate head maps (v_i, u_i) to the silence logit. A softmax over the $1+K|\mathcal{P}|$ outputs defines π_θ. This set aggregation follows the permutation-invariant construction of Deep Sets [14]; permuting the subchannel histories permutes their action logits. The GRU state is updated during idle slots and retained across packet boundaries, then reset when a new training or evaluation episode begins. Empty devices are masked to OFF.

B. Energy reward and constraint handling

A centralized state-value critic observes all devices' features and true gains during training. Let $c_i(t)$ be the deadline-violation indicator settled at the end of slot t. The common training reward is

$$r_t = -\frac{1}{N}\left(\sum_i \frac{e_i(t)}{E_{\mathrm{ref}}} + \sum_i \lambda_i(t) c_i(t)\right), \quad E_{\mathrm{ref}} = \tau p_{\max}. \tag{9}$$

Our primary method, *MAPPO-F*, sets $\lambda_i = w = 10$ for all devices. The penalty weight is selected on validation data.

The comparison method, *MAPPO-D*, maintains a multiplier locally at each device. At each of that device's deadlines,

$$\lambda_i \leftarrow \mathrm{clip}\big(\lambda_i + \eta(c_i - \epsilon), 0, \lambda_{\max}\big). \tag{10}$$

The multiplier is also an actor input. We use $\eta = 2$ and $\lambda_{\max} = 50$, independent uniform initial multipliers on $[0, 50]$ during training, and a common initial value selected from $\{0, 5, 25\}$ during validation. Execution updates depend on local outcomes, implementing state augmentation with dual dynamics [12].

Both variants use the clipped MAPPO objective [11], with a common generalized advantage estimate from the centralized critic and an entropy bonus. Only devices with pending packets contribute to the policy loss and entropy; the critic learns from all slots. Ordered trajectory replay preserves recurrent state. At technical episode truncations, the critic bootstraps the value of the continuing communication task.

V. Experimental Evaluation

A. Scenario and evaluation protocol

Table I gives the scenario and training parameters. Positions are uniform by area in a disk, with gains $g_i =$

TABLE I
SCENARIO AND MAPPO PARAMETERS.

Parameter	Value
Devices / subchannels	$N \in \{4, 6, 8, 12\}$ / $K = 4$
Slot / bandwidth per subchannel	1 ms / 1 MHz
Cell radius / path-loss exponent	100 m / 4
Noise density / noise figure	-174 dBm/Hz / 7 dB
Positive powers	5, 10, 15, 19, 23 dBm
Periods (slots)	10, 20, 40
Packet sizes (bits)	6144, 12 288, 24 576
Violation allowance	$\epsilon = 0.05$
Evaluation horizon / warm-up	2000 / 200 slots
Training episode / rollout	600 / 128 slots
Environments / PPO epochs	16 / 4
Actor GRU / critic widths	64 / 256, 256
Actor / critic learning rates	3×10^{-4} / 10^{-3}
Discount / GAE parameter	0.99 / 0.95
PPO clip / gradient norm limit	0.2 / 0.5
Budget per seed	20×10^{6} environment slots

$10^{-4}\max(d_i/1\,\mathrm{m}, 1)^{-4}$, and remain fixed within an episode. Periods and packet sizes give equal offered load per device. Traffic-class counts for $N = 4, 6, 8, 12$ are $(2, 1, 1)$, $(2, 2, 2)$, $(3, 3, 2)$, and $(4, 4, 4)$, respectively.

For each N, we train separate policies with seeds 101, 202, and 303, drawing each episode from one million training instances. Development, training, validation, test, and confirmation pools are disjoint. The study assesses generalization to new topologies and phases at a trained N. The MAPPO entropy coefficient decreases from 0.01 to zero over ten million slots; minibatches contain four environment trajectories. The penalty and dual step size were selected in a two-million-slot study at $N = 8$ and transferred to the long runs.

Validation uses 64 instances, one stochastic execution each, every million training slots. An episode is feasible when all devices satisfy $V_i \le \lfloor 0.05 M_i \rfloor$, where M_i and V_i count evaluated packets and violations. Let F be the fraction of feasible episodes. Among checkpoints with validation $F \ge 0.95$, selection minimizes mean E/U_0; otherwise it maximizes F, breaking ties by energy. MAPPO-D jointly selects its initial multiplier. Both selected and final checkpoints are evaluated, with selected checkpoints providing the main results.

Testing uses 200 unseen instances per N, two stochastic executions each, and all three training seeds. Private device random streams use common seed assignments across methods. Evaluated packets have deadlines in $(200, 2000]$; E includes their full lifetime transmit energy, including failed packets, matching the centralized packet set. Mean power instead counts transmissions in slots $[200, 2000)$. A method meets the feasibility criterion when the point estimate of aggregate F is at least 0.95. We compare energy among methods meeting this criterion and calculate percentage savings from their mean normalized energies. We use 10 000 two-level bootstrap resamples of training seeds and instances, averaging repetitions within an instance. Method differences resample instances jointly and training seeds independently. All reported intervals are 95% percentile intervals.

TABLE II
TEST-SET RESULTS: F / MEAN E/U_0. † MARKS $F < 0.95$.

Method	$N = 4$	$N = 6$	$N = 8$	$N = 12$
MAPPO-F	0.993/1.01	0.986/1.09	0.994/1.21	0.984/1.62
MAPPO-D	0.992/1.01	0.993/1.11	0.979/1.46	0.972/1.52
Paced-sticky	1.000/1.00	0.990/1.73	0.960/2.80	0.980/5.51
Random-max	1.000/9.87	1.000/9.32	1.000/9.34	0.995/8.93
Sticky-max	1.000/10.36	1.000/10.05	1.000/10.40	0.983/10.21
Lazy-pacing	0.797/5.34†	0.657/4.98†	0.287/5.76†	0.000/8.02†

B. Local rules and held-out performance

Random-max sends on a random subchannel at maximum power. *Sticky-max* retains its subchannel when previous service meets the current pace s_i/D_i, otherwise reselecting randomly, at maximum power. *Lazy-pacing* uses its latest link estimate to choose the lowest power meeting that pace and changes subchannel when necessary. The strongest feasible rule, *paced-sticky*, remembers its gain across packets and retains a subchannel within 3 dB of the minimum interference. Otherwise it switches with probability 0.5 to one within 3 dB of that minimum. It selects the lowest power predicted to serve $(1 + m)s_i/D_i$ bits, probing at maximum power until its gain is known. Validation calibration over $m \in \{0, 0.1, 0.25, 0.5, 1, 2, 4, 8, 16, 32\}$ selects $0, 0.25, 0.5, 8$ for the four loads. All rules obey the same execution information boundary.

MAPPO-F meets the feasibility criterion at all four loads (Fig. 3, Table II). For $N = 6, 8, 12$, its mean E/U_0 is 1.094, 1.213, and 1.623, versus 1.725, 2.798, and 5.508 for paced-sticky: reductions of 36.6%, 56.7%, and 70.5% in mean normalized energy. Paired MAPPO-F minus paced-sticky energy intervals are $[-0.659, -0.603]$, $[-1.632, -1.541]$, and $[-4.179, -3.535]$. Mean powers are 4.36, 5.30, and 8.87 mW/device, versus 6.83, 12.07, and 29.24 mW/device.

At $N = 4$, paced-sticky attains the exact zero-loss reference; MAPPO-F gives $E/U_0 = 1.007$ $[1.004, 1.010]$ and $F = 0.993$. At $N = 6, 8, 12$, MAPPO-F's mean E/L_0 is 1.119, 1.349, and 2.289. Thus substantial distance from centralized scheduling remains at high load. Maximum-power rules are reliable but expensive; lazy-pacing fails the feasibility criterion at every load.

C. Comparison with an XR-SBPC-style learner

We adapt the discrete soft actor–critic (SAC) scheme of XR-SBPC [10] to the same environment and actor features. Shared feedforward actor and twin local Q networks use widths 400/300/200/100, LeakyReLU in the first three layers and tanh in the last. The learning rate is 10^{-4}, batch size 512, replay capacity 10 000, target update rate 0.005, and target entropy 0.05, with initial temperature 1. Proportional prioritized replay uses exponent 0.6 and importance exponent annealed from 0.4 to 1. One update follows each step of 16 environments, starting when the buffer is full. We evaluate discounts of 0.5 and 0.99.

Three rewards distinguish the objective from the training scheme. *SAC-V1* gives a device $+50$ on completion and -100

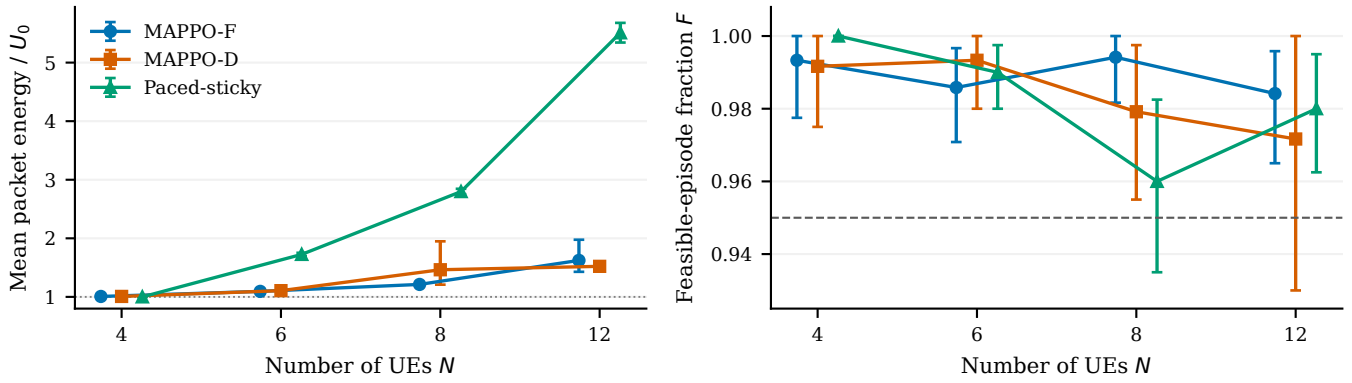


Fig. 3. Selected checkpoints on the test set: mean normalized energy and feasible-episode fraction, with 95% bootstrap intervals. U_0 is a centralized zero-loss witness, exact for $N = 4$; $F = 0.95$ is the feasibility threshold.

TABLE III
SELECTED CHECKPOINTS ON CONFIRMATION SET C1, $N = 8$. † MARKS $F < 0.95$.

Method	γ	F	Mean E/U_0 [95% CI]
MAPPO-F	0.99	0.993	1.217 [1.209, 1.225]
MAPPO-D	0.99	0.979	1.467 [1.214, 1.949]
Paced-sticky	–	0.950	2.821 [2.772, 2.870]
SAC-V1	0.99	0.979	7.101 [5.876, 8.669]
SAC-V1s	0.99	0.988	7.487 [5.719, 8.786]
SAC-V2	0.99	0.978	3.028 [2.863, 3.212]
SAC-V1	0.5	0.988	7.222 [6.505, 7.899]
SAC-V1s	0.5	1.000	7.655 [7.204, 8.161]
SAC-V2†	0.5	0.621	5.390 [5.232, 5.561]

on expiry. Otherwise, while a packet is pending, it receives $5b_i/(s_i/D_i)$ if $b_i \geq s_i/D_i$, and -10 if below that pace. Idle reward is zero. *SAC-V1s* retains only completion and expiry rewards. *SAC-V2* uses exactly MAPPO-F's team reward (9), including on idle transitions. All variants train Q values on idle transitions, but exclude them from actor and temperature losses, avoiding bootstrap from untrained idle states. Neighbor actions and instantaneous channel gains available in the original study are omitted to match our execution information.

All 3 rewards $\times$ 2 discounts $\times$ 3 seeds receive the same 20-million-slot budget as MAPPO. Following the cross-load tests, we compare the frozen policies on an independent confirmation set, C1: 200 new instances, two executions each, with separately computed references. Checkpoint selection and analysis rules were fixed before C1 evaluation; the existing MAPPO policies were frozen. Reward comparisons fix $\gamma = 0.99$ in advance. For the comparison of complete schemes, validation on seed 101 selects one discount per reward, applied to all three seeds: 0.5 for V1/V1s and 0.99 for V2.

Table III reproduces the earlier ranking: MAPPO-F uses $E/U_0 = 1.217$, versus 2.821 for paced-sticky, a 56.9% reduction. Paced-sticky meets the criterion exactly ($380/400$ feasible executions); its F interval is $[0.922, 0.973]$. At fixed $\gamma = 0.99$, both reliability rewards and V2 meet the criterion. V1 minus V2 and V1s minus V2 energy differences are 4.073 [2.838, 5.634] and 4.459 [2.689, 5.737], respectively. Switching from individual reliability rewards to the energy-aware team reward significantly reduces energy at $\gamma = 0.99$. V1 minus V1s has interval $[-2.347, 1.651]$, giving no resolved energy effect of shaping. At $\gamma = 0.5$, V2 fails the feasibility criterion; selection returns its untrained initial checkpoint.

With the same energy reward and execution information, V2 minus MAPPO-F has energy difference 1.811 [1.648, 1.999], corresponding to 59.8% lower normalized energy for MAPPO-F. Their feasibility difference interval, $[-0.036, 0.007]$, includes zero. V2's selected checkpoints occur at 3–7 million slots, while its final F falls to 0.636 on C1.

D. Access behavior and reliability

Selected MAPPO-F and MAPPO-D satisfy the feasibility criterion at every N, but their paired energy intervals all contain zero. At $N = 8$, means are 1.213 and 1.463, with difference interval $[-0.737, 0.005]$; C1 gives $[-0.734, 0.003]$. Final MAPPO-F checkpoints remain feasible at all loads ($F = 0.987$–0.992), whereas final MAPPO-D drops to 0.915 and 0.938 at $N = 8, 12$. The fixed penalty is therefore our simpler primary configuration.

Co-channel overlap is the fraction of transmissions sharing a subchannel with another active device. Interference loss is one minus total actual mutual information divided by the interference-free service of the same transmissions. For MAPPO-F, short (2M) and long (20M) configurations reduce overlap from 0.712 to 0.211 and loss from 0.418 to 0.064 (Fig. 4). Their test energies are 2.321 and 1.213. The long configuration extends both training and entropy annealing. Explicit interference histories at lags 10/20/40 did not reach a predeclared 5% improvement threshold in a separate short-budget validation experiment.

At $\gamma = 0.99$, SAC-V2 sends in nearly every slot with a pending packet, with overlap 0.58–0.68 and interference loss 0.38–0.40 across seeds. MAPPO-F's corresponding conditional transmission fraction is 0.61–0.76; 82–85% of its transmissions use 5 or 10 dBm. V1/V1s instead have mean

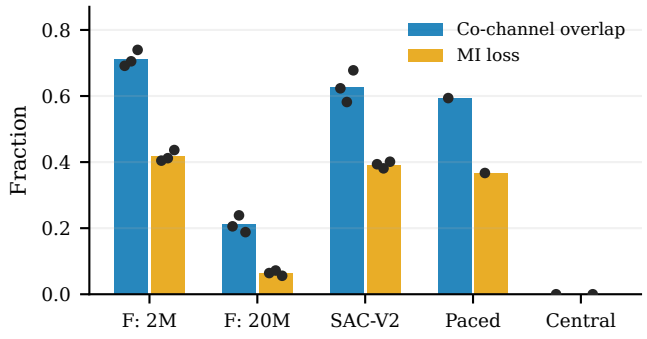


Fig. 4. Validation diagnostics at $N = 8$. F denotes MAPPO-F; SAC-V2 uses $\gamma = 0.99$. Bars are seed means and dots are individual seed summaries. The local rule and central witness each have one summary.

completion times of 0.16–0.27 of a period, often transmitting at high power. MAPPO-F combines low-power service with selective silence. No SAC run beats paced-sticky's validation energy while meeting $F \geq 0.95$; MAPPO-F first does so after 1–4 million slots.

Pooling selected MAPPO-F/D over four loads gives 128 infeasible executions out of 9600. The worst device is 75–100 m from the AP in 108 of these, and has period 40 in 83. Such devices have only 45 evaluated packets and can miss at most two. A MAPPO-F execution at $N = 4$ reaches device violation rate 0.733.

E. Scope and limitations

Experiments use static, frequency-flat channels, exact feedback, and ideal accumulated service, and count radiated transmit energy. Fading, finite-blocklength coding, coding gaps, and feedback, circuit, and processing costs remain extensions. The finite-episode feasibility criterion provides empirical reliability evidence, without a per-instance constraint guarantee. Centralized energy references impose zero loss; their ratios do not bound optimality under the 5% violation allowance. Results use three training seeds, and the XR-SBPC-style adaptation covers $N = 8$ and two discounts with transferred hyperparameters. The learning comparison assesses complete training schemes. Controlled ablations are needed to distinguish architectural and training contributions, the roles of power adaptation and silence, and the joint effects of training budget and entropy scheduling.

VI. Conclusion

A shared recurrent MAPPO controller learns energy-efficient uplink access from local packet state and delayed AP feedback. With 6–12 devices sharing four subchannels, MAPPO-F reduces mean normalized transmit energy by 36.6%–70.5% relative to calibrated paced-sticky access, while meeting the per-device feasibility criterion. On the independent eight-device confirmation set, the reduction reaches 59.8% against an XR-SBPC-style SAC adaptation with the same energy reward and execution information. Centralized zero-loss references quantify the remaining energy gap. The learned policies combine low-power transmission and selective silence with substantially reduced co-channel interference.

References

[1] E. Uysal-Biyikoglu, B. Prabhakar, and A. El Gamal, "Energy-efficient packet transmission over a wireless link," *IEEE/ACM Transactions on Networking*, vol. 10, no. 4, pp. 487–499, 2002.

[2] E. Uysal-Biyikoglu and A. El Gamal, "On adaptive transmission for energy efficiency in wireless data networks," *IEEE Transactions on Information Theory*, vol. 50, no. 12, pp. 3081–3094, 2004.

[3] I.-H. Hou and P. R. Kumar, "Scheduling heterogeneous real-time traffic over fading wireless channels," in *Proc. IEEE INFOCOM*, 2010.

[4] L. Deng, C.-C. Wang, M. Chen, and S. Zhao, "Timely wireless flows with general traffic patterns: Capacity region and scheduling algorithms," *IEEE/ACM Transactions on Networking*, vol. 25, no. 6, pp. 3473–3486, 2017.

[5] J. Bai, Z. Chen, and E. G. Larsson, "Delay-constrained grant-free random access in MIMO systems: Distributed pilot allocation and power control," *IEEE Transactions on Cognitive Communications and Networking*, vol. 11, pp. 1535–1550, 2025.

[6] O. Naparstek and K. Cohen, "Deep multi-user reinforcement learning for distributed dynamic spectrum access," *IEEE Transactions on Wireless Communications*, vol. 18, no. 1, pp. 310–323, 2019.

[7] L. Liang, H. Ye, and G. Y. Li, "Spectrum sharing in vehicular networks based on multi-agent reinforcement learning," *IEEE Journal on Selected Areas in Communications*, vol. 37, no. 10, pp. 2282–2292, 2019.

[8] C. Guo, Z. Li, L. Liang, and G. Y. Li, "Reinforcement learning-based power control for reliable mission-critical wireless transmission," *IEEE Internet of Things Journal*, vol. 10, no. 23, pp. 20 868–20 883, 2023.

[9] Y. Zhang and D. Guo, "Multi-agent reinforcement learning for multi-cell spectrum and power allocation," *IEEE Transactions on Communications*, vol. 73, no. 8, pp. 5980–5992, 2025.

[10] S. Bagherinejad, T. Jacobsen, N. K. Pratas, and R. O. Adeogun, "DRL-based distributed joint sub-band allocation and power control for eXtended Reality over in-body subnetworks," in *Proc. IEEE Wireless Communications and Networking Conference (WCNC)*, 2025.

[11] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu, "The surprising effectiveness of PPO in cooperative multi-agent games," in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 24 611–24 624.

[12] M. Calvo-Fullana, S. Paternain, L. F. O. Chamon, and A. Ribeiro, "State augmented constrained reinforcement learning: Overcoming the limitations of learning with rewards," *IEEE Transactions on Automatic Control*, vol. 69, no. 7, pp. 4275–4290, 2024.

[13] G. Caire and D. Tuninetti, "The throughput of hybrid-ARQ protocols for the Gaussian collision channel," *IEEE Transactions on Information Theory*, vol. 47, no. 5, pp. 1971–1988, 2001.

[14] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Póczos, R. Salakhutdinov, and A. J. Smola, "Deep sets," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.